

CSET418– ADVANCED FRONTEND DEVELOPMENT

Course Type -Specialization Core L-T-P Format 6-0-4 Credits– 4

The course covers **Advanced Front-End Development, Client-server Architecture, Front end programming tools, and library**. This includes basics of **Advanced JS and React**. We will be doing 2 Advanced Projects along with that.

COURSE-SPECIFIC LEARNING OUTCOMES

CO1: To understand the Client-Server architecture and the basics of client-side programming languages and frameworks.

CO2: To examine of client-side libraries: JS in Browser, JS Asynchronous Programming, React, React DOM, React-Router DOM, Redux Libraries.

CO3: To implement client-side application development.

DETAILED SYLLABUS

Module 1 (W1, W2, W3, W4, W5, W8, W9, W10) JavaScript in Browser-Introduction, Chrome Foundation Services, Runtime Environment, Memories, Browser Initialisation Process, Browser Rendering Engine, V8, Engine, Execution Context, Asynchronous JS, Callback, Promises, Async, Await, Web API's Introduction, Forms, History, Storage, Worker, Fetch, Geolocation.

Module 2 (W11, W12, W13, W18, W19, W20, W21) Introduction to React JS, React Virtual DOM, React JSX, React Components, React Props, React Conditionals, Custom React, React State, React Hooks, React State Management, React Redux, React Router DOM.

STUDIO WORK / PROJECTS

Module 1(W1, W2, W3, W4, W5, W8, W9, W10) Project-1(HTML, CSS, JS, Juvizo).

Module 2 (W11, W12, W13, W18, W19, W20, W21) Project-2(HTML, CSS, React, MeTube).

EXAMINATIONS

Mid-Term and Syllabus (W6, W7) Module-1.

End-Term and Syllabus (W14, W15, W16, W17) Complete Syllabus.

